

Response to Reviewers

The Energy Cost of Privacy in Federated Learning
cSIS 2026 — Track 1, Green & Intelligent Computing

We thank both reviewers for their careful reading. The second reviewer’s comments led us to re-derive several numbers directly from the raw logs; the most consequential outcome is a correction to the privacy guarantee (Point 3 of Reviewer 2), which we now report as a finding in its own right. All tables, figures and in-text numbers in the revision are generated by a single analysis script from one measurement sweep per dataset, and no figure has been edited by hand.

Reviewer #1

We thank the reviewer for the accurate summary of our contributions and for recognizing the experimental controls, which were central to the study’s design.

Page limit. The revision fits within six pages. The space came from removing duplication between Sections IV and V and from shortening the captions; the carbon figure was also dropped, since Table III carries the same six numbers. No experimental result was removed for space.

Keywords. The heading now reads “Keywords”.

Title. The title is now set in bold.

Captions. All figure and table captions now state only what is shown, the units and the averaging. Interpretive material has moved into the body text.

Reviewer #2

1. Two baselines and two noise floors. We thank the reviewer for identifying this. The submitted manuscript combined two measurements of the non-private condition: the randomized sweep that all other conditions come from, and a separate later re-run of that condition alone, used in Fig. 2 and the opening of Section IV. Only the sweep survives — the re-run’s files were overwritten by a subsequent pass and cannot be regenerated — so we can defend only the sweep, and the revision uses it throughout. The two differ by 3.7%, inside the noise floor, so nothing downstream changes qualitatively. Every number in Tables I–III, Figs. 1–2 and the text now comes from that sweep: baseline 13.98 J/round, mean overhead +51%, range +43.4% to +62.2%, board-power noise floor 7.80%. The 14.5 J / +45% / 8.2% / 9.46% set is withdrawn.

On ε -dependence we also changed the instrument, because the submitted comparison was mismatched twice over. The 12.5% spread is a spread in net energy per round, while the 7.80% floor is a spread in board power; and a floor bounds the seeds of *one* condition, whereas the claim concerns the spread of *five condition means*. Each floor is now measured on the quantity it bounds — on energy per round it is 12.0%, not 7.8% — and the ordering claim is tested rather than compared to a floor: permuting the seed labels across the five private budgets gives $F = 1.46$, $p = 0.27$ on HAR and $F = 0.46$, $p = 0.79$ on CIFAR-10. The contrast the reviewer raises, +43.4% at $\varepsilon = 1$ against +62.2% at $\varepsilon = 4$, is not resolvable: the three seeds of the $\varepsilon = 4$ condition span 20.9 to 24.1 J/round on their own, and the overheads are not ordered in ε . Section IV-A is retitled accordingly and claims no resolvable dependence on ε , and explicitly no equality. Across budgets, per-round energy varies by at most $1.13\times$, against $1.28\times$ to unbounded for the round-count term.

2. Numbers not reproducible from the tables. The 0.50-target range, the 0.70 ratio and the 0.85 entry in the submitted text were stale and have been regenerated ($1.28\text{--}1.79\times$, $5.7\times$, 839 J over 57.0 rounds). The per-round discrepancies the reviewer identifies had a different cause: the 2477 J post-peak figure and the 29 J / 64 J ablation figures were gross energy, while Table I reports net of idle. Gross energy at $\varepsilon = 0.5$ is 28.8 J/round, which over 83 rounds gives ≈ 2390 J; the two ablation cells are 30.5 and 64.7 J/round gross against 22.0 and 48.6 net. The revision reports every energy figure net of idle, labels any gross figure explicitly, and generates all in-text numbers from the same arrays as the tables. We also note in Section IV-E that the ablation sweep’s own baseline sits about 2 J/round above Table I’s, so ablation energies are compared only within that sweep.

3. Privacy guarantee and step count. We agree on both points. (a) Per-sample clipping and noising within each client yields example-level DP, enforced locally against an untrusted server and accounted per client over the whole run; it is not client-level DP. On HAR it protects individual windows of a participant’s data, not the participant’s participation. The guarantee is relabelled throughout (abstract, Sections I, II-D, III-B, V-E, conclusion), and [2] and [11] are now cited as client-level mechanisms, stronger than ours; the per-round overhead we measure is the

per-sample gradient work that both need, so the energy result is orthogonal to the choice between them. (b) The reviewer is right that calibrating σ on expected participation does not bound the realized guarantee. We recomputed the realized ε of every client from its logged participation with the same RDP accountant. Clients joined between 33 and 64 rounds against an expected 50, and the worst realized ε exceeds the nominal budget in every private run — by 4–14% on HAR (0.570 for nominal 0.5, 9.11 for nominal 8) and by up to 22% on CIFAR-10. Table I now reports the worst realized ε per condition, the sweep is described as a sweep over nominal budgets, and the claim that the reported ε is an upper bound has been removed; Section III-B states the remedy, which is to calibrate against $RE[n/B]$ or cap participation per client, and the audit script reports that bound directly. The energy measurements are unaffected, since they depend only on σ . We now present this as a second accounting hazard alongside the calibration-timing hazard: both are silent in the resulting numbers, and only a replay of the logs exposes either.

4. The 83% and $21.4\times$ figures. We agree, and a per-seed analysis prompted by this comment goes further. On HAR at $\varepsilon = 0.5$ the three seeds peak at rounds 16, 54 and 65, and the mean curve shows no early peak. The HAR stopping-round claim is therefore not supported and has been withdrawn, together with the 83% figure, which, as the reviewer notes, restated the schedule. On CIFAR-10 the per-seed check does support the effect, but only at the two tightest budgets: at $\varepsilon = 0.5$ the decline is 0.026 against a ± 0.010 across-seed spread with the seeds peaking at rounds 8, 8 and 9, and at $\varepsilon = 1$ it is 0.044 against ± 0.014 with the seeds between rounds 10 and 16. At $\varepsilon \geq 2$ the declines fall inside their spreads. We therefore retain the stopping round on CIFAR-10 at $\varepsilon \leq 1$ and withdraw the monotone ordering of the peak round, which only two budgets resolve. Section IV-C now leads with the peak’s position rather than the share of joules after it, and reports a peak only where the decline clears that condition’s spread and the seeds agree on where it falls. We also introduce a separate accuracy noise floor, since the energy floor had been applied to accuracy differences. On carbon, we agree that $21.4\times$ is the grid-intensity ratio and would apply equally to non-private training. It has been removed from the abstract, and Section IV-F now presents per-grid carbon attributable to privacy as context for federated deployment, where no operator chooses the grid.

5. References. We thank the reviewer for the verification. The truncation to one author was the bibliography style’s default name limit rather than missing data; the style now lists up to six authors before “et al.” as IEEE requires. Volume, issue and page numbers have been added where they exist, and the DOI for Aroussi et al. is 10.1109/IRASET68627.2026.11538879. The entries that lacked volume and pages were preprints typed as journal articles; they are now identified as preprints with their arXiv identifiers, so no entry claims a volume it does not have, and two IEEE entries are marked early access. We were unable to confirm archival versions for the remaining preprints (Opacus, Flower, and the Green-AI estimation tools); if the reviewer has a specific published version in mind for [12], we will substitute it. Anguita et al. carried a spurious “and others” — the paper has five authors — and is now cited by its ESANN paper number, since we could not confirm against the proceedings the page range that circulates in secondary sources.

6. Page limit. The revision is six pages. Sections V-A and V-B have been condensed to the implications not already stated in Section IV, and the measurement-hazard discussion (Section V-D) is preserved in full and extended with the participation hazard of Point 3.